



Online education gets a boost

Struggling online learning networks have gotten a shot in the arm thanks to coronavirus. <https://digit.in/jun20-92>



WikiHow's cheap art

WikiHow has used cheap labour to create custom artwork and rise in search engine rankings. <https://digit.in/jun20-93>

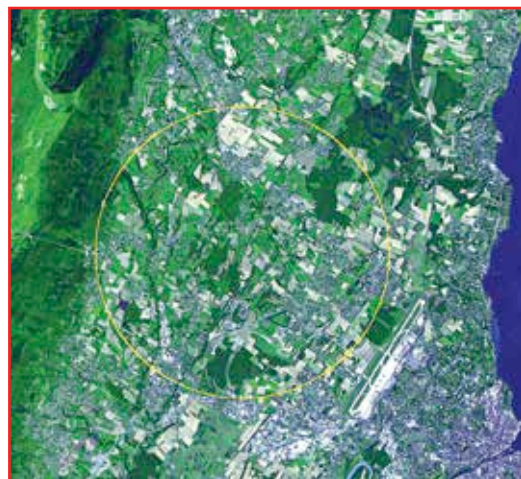
is that the LHC has also identified and confirmed a few other theories as well. One of the first major discoveries of the LHC was to confirm the existence of tetraquarks. We don't have space to explain those here, but it got theoretical physicists jumping up and down, which should be good enough for us to accept that it was a big deal.

When scientists at the LHC got bored of colliding protons, they decided to bash protons into the nucleus of lead. This ended up creating a truly exotic state of matter called a quark-gluon-plasma (QGP), which was previously only created by colliding lead and gold. This new way of creating QGP has scientists scratching their collective heads. We're only being flippant for effect – no one is bored at the LHC, no one decides to just smash new stuff together without obtaining permission and a consensus that it's a good use of precious collider time, but because of how cutting edge the science is, we're pretty sure there's more head scratching at the LHC than a kindergarten class infested with lice!

The most interesting thing about the LHC is what it hasn't yet discovered, which is supersymmetry, also called SuSy. Again, we don't have the space to explain that here, but you can read more about the LHC in our mystify LHC, May 2018 issue.

OPEN SOURCE

Many people aren't aware that it is not paid software that was the norm



A NASA image that shows the scale of the LHC

early on, but actually open source. The problem is that most software history lessons start with the Apple-Microsoft competition to try and dominate the desktop. In fact, software was much older, albeit much simpler, all the way from the 1950s. Most software was developed specifically for the hardware running it, and without set standards, this meant that a lot of it was nice. Early software that people dabbled with was almost exclusively academic, and developed in universities. Sharing source code was actually the norm, because of the aforementioned lack of hardware standards.

When computers started getting more complex, and became more powerful in the early 1970s, the time and effort to develop software started going up, which was directly associated with costs. Plus, the advantages of commercial use of computers was becoming more obvious to everyone. When software was made copyrightable in 1974 in the US is when the paid software phenomenon truly began. Greed took over and everything started becoming copyrighted. Enter Richard Stallman, and started the GNU (GNU's not Unix) Project. He invented the phrase Copyleft, and started the Free Software Foundation, thanks to which so much collaboration across the globe happens today. It's not a stretch to claim that the very internet itself is far more dependent on free and open source software (FOSS) than paid software. So many tools that we can think of – Android, Chrome, Firefox, Linux, PHP, MySQL, Wordpress, Apache, etc. – is all FOSS. Arguably, in the technology world, FOSS is mankind's greatest collective achievement!



Richard Stallman, hero to FOSS lovers everywhere!

OTHERS

Honestly, it would take dozens of pages and tens of thousands of words to list out everything that we think of as definitive to our technological success. Since that's beyond the scope of this article, we thought we'd list out the few that we really wanted to include, and see if you

agree with them, or if you have others that you think we missed out on.

- **Encryption** – because everything we do online today would be impossible without it!
- **3D Printing** – Also called additive manufacturing, this is something that has allowed us to completely rethink how we design things, and will play a huge role in our future as well.
- **Social web** – The jury is still out, is it a great achievement or our biggest failure? Will it result in a global society, or divide us up into groups and bring down society? We don't know, but good or bad, it's certainly something we cannot ignore!
- **Cheap storage** – Perhaps a subset of Moore's Law, if we took a very loose definition, storage has progressively gotten cheaper per GB, to the point where everyone is only talking TBs and some are starting to talk about PBs of storage!
- **AI / ML / NLP** – While we're all waiting for General AI from sci-fi to come along, dumb AI has brute forced its way into every aspect of our lives. From the phones we carry to the heart of all the services we use, AI, machine learning or natural language processing is improving at an impressive rate!
- **YouTube?** – Is there anyone who doesn't use the internet to watch videos any more? If you're looking to experience a new gadget, waste some time with cat videos, or learn a new skill, we're doing almost all of it through video. YouTube has to be given considerable credit for kickstarting that! 📺